

LAMIA, Greece

WMO: 166750

Lat: 38.8767N

Lon: 22.4361E

Elev: 12

StdP: 101.18

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	-0.2	-7.5	2.0	3.2	-5.3	2.4	3.5	10.5	11.6	8.9	8.2	2.7	290	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	35.6	20.9	34.2	20.8	33.1	20.4	22.5	31.3	21.9	31.1	21.4	30.3	4.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	14.8	24.5	19.4	14.1	24.4	18.7	13.5	24.2	66.1	30.8	64.1	31.1	62.1	30.1	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.2	(b) 7.7	(c) 6.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	7.6	8.2	10.5	14.6	19.5	24.4	25.6	25.6	22.5	17.4	12.7	9.0		
	DBStd	7.28	3.08	3.43	3.27	2.92	3.03	2.65	2.16	1.85	2.78	3.31	3.46	3.05		
	HDD10.0	257	84	67	35	2	0	0	0	0	0	0	15	54		
	HDD18.3	1519	331	283	244	116	24	1	0	0	4	56	171	291		
	CDD10.0	2637	11	18	50	141	293	431	485	485	374	231	96	22		
	CDD18.3	856	0	0	0	4	59	182	227	227	128	28	2	0		
	CDH23.3	9071	0	0	1	63	597	2033	2554	2395	1200	218	10	0		
	CDH26.7	4132	0	0	0	11	169	929	1301	1202	474	45	1	0		
(14) Wind	WSAvg	2.6	2.4	2.6	2.7	2.9	3.0	3.1	2.8	2.7	2.7	2.2	2.1	2.3		
(15) Precipitation	PrecAvg	523	58	52	54	35	37	16	18	18	32	65	73	66		
	PrecMax	737	147	104	127	93	72	73	67	85	92	241	256	153		
	PrecMin	354	5	4	18	3	6	0	0	0	0	3	9	1		
	PrecStd	86	37	29	29	23	21	15	18	22	24	49	53	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	20.4	22.8	28.4	32.4	37.0	37.7	36.5	34.9	30.3	24.4	20.2		
		MCWB	11.5	13.0	13.4	16.2	19.2	20.8	21.8	21.1	20.1	18.6	15.4	14.0		
	2%	DB	16.0	18.2	20.8	25.2	30.1	34.8	35.6	35.0	32.6	27.9	21.9	17.7		
		MCWB	11.0	12.0	12.9	15.2	18.1	20.4	21.1	21.1	19.9	18.2	15.1	12.6		
	5%	DB	14.4	16.5	19.2	23.0	28.5	33.2	34.1	33.7	31.1	25.7	20.1	15.4		
		MCWB	10.4	11.2	12.2	14.6	17.8	20.2	21.1	20.7	19.3	17.5	14.5	11.1		
	10%	DB	13.0	14.7	17.4	21.2	26.8	31.7	32.8	32.6	29.7	24.0	18.2	13.9		
		MCWB	9.9	10.2	11.4	13.8	17.3	19.8	20.7	20.5	19.0	16.9	14.0	10.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.9	14.9	17.0	20.7	22.7	23.2	23.2	22.3	20.4	17.4	14.8		
		MCDB	15.7	18.7	20.3	24.8	29.0	32.2	32.2	31.0	29.1	26.5	20.7	18.2		
	2%	WB	12.1	12.7	13.7	16.2	19.4	21.7	22.5	22.6	21.2	19.3	16.2	13.0		
		MCDB	14.8	16.9	18.8	23.5	27.7	32.0	32.1	30.7	28.8	24.8	19.7	16.6		
	5%	WB	11.1	11.8	12.8	15.4	18.6	21.1	21.9	21.9	20.5	18.5	15.3	11.9		
		MCDB	13.4	15.7	18.1	21.6	26.2	30.8	31.7	30.2	28.1	23.6	18.8	14.6		
	10%	WB	10.2	10.6	11.9	14.5	17.8	20.5	21.3	21.4	19.9	17.7	14.5	10.9		
		MCDB	12.5	14.0	16.6	19.9	24.9	29.8	30.6	29.6	27.4	22.5	17.8	13.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	9.8	11.3	11.7	13.2	13.7	13.5	13.0	10.3	9.6	8.6		
		MCDBR	11.6	13.3	13.9	14.4	14.6	15.3	15.9	15.1	15.5	14.0	12.4	11.5		
		MCWBR	7.3	7.8	7.3	6.5	5.8	5.1	4.9	5.0	5.3	5.9	6.4	7.0		
	5% WB	MCDBR	9.0	11.7	12.4	13.6	13.5	13.7	14.2	13.1	13.8	11.7	11.0	9.8		
		MCWBR	6.0	7.2	7.1	6.6	5.9	5.2	4.8	5.0	5.5	5.5	6.1	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.377	0.412	0.464	0.518	0.503	0.483	0.491	0.502	0.480	0.445	0.410	0.384		
	taud		2.308	2.168	2.026	1.910	2.011	2.116	2.103	2.065	2.102	2.166	2.221	2.266		
	Ebn at Noon		775	792	786	769	791	806	795	775	764	746	724	729		
	Edh at Noon		95	124	158	188	174	156	157	160	145	123	103	93		
(44) All-Sky Solar Radiation	RadAvg		1.93	2.68	3.90	5.15	6.15	7.05	7.32	6.55	4.82	3.26	2.24	1.70		
	RadStd		0.28	0.31	0.36	0.48	0.33	0.46	0.33	0.27	0.35	0.34	0.23	0.26		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.35	N/A	N/A	N/A	+0.62	N/A	N/A	N/A	-83	+97	N/A

Nomenclature: See separate page